

# Normalizing Flows

GANs: can't compute  $p(x)$ , unstable  
discr./gen. training

NFs: can compute  $p(x)$ , stable max  
likelihood training

Anomaly detection: if  $p(x_{\text{test}}) < \epsilon$ , then  $x_{\text{test}}$  outlier

Compare two NFs / tune hyperparameters by  
$$\sum_i \log p(x_{\text{test}, i})$$

$$z \sim p(z) = \mathcal{N}(z|0, I)$$

$x = f(z, \theta)$ ,  $f$  - invertible one-to-one transform

$$z = f^{-1}(x, \theta)$$

$$p_X(x) = p_Z(\underbrace{f^{-1}(x)}_z) \left| \det \frac{\partial f^{-1}(x)}{\partial x} \right| =$$

jacobian  $\dim x \times \dim x$

$$= p_Z(z) \left| \det \frac{\partial f(z)}{\partial z} \right|^{-1}$$

Example  $z \sim N(z | \mu, \sigma^2)$

$$x = \exp(z) \Rightarrow z = \log x$$

$$p_X(x) = N(\log x | \mu, \sigma^2) \left| \det \frac{\partial \log x}{\partial x} \right| =$$

$$= \frac{1}{x} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right) = N(z | \mu, \sigma^2) \left| \det \frac{\partial \exp(z)}{\partial z} \right|^{-1}$$

$\exp(z) = x$

Flow:  $z_0 \sim p(z)$ ,  $z_1 = f_1(z_0, \theta_1)$ ,  $z_2 = f_2(z_1, \theta_2), \dots,$

$$x = z_k = f_k(z_{k-1}, \theta_k)$$

### NF scheme

Choose  $f_k(z, \theta) \forall k=1, \dots, K$

Init.  $\theta$

Until convergence

$x_1, x_2, \dots, x_M \sim p_{\text{data}}(x)$

for each  $x_i$ :

$$z_k^i = x_i, z_{k-1}^i = f_k^{-1}(z_k^i, \theta), \dots, z_0^i = f_1^{-1}(z_1^i, \theta)$$

$$\text{Loss} = -\frac{1}{M} \sum_{i=1}^M \left( \log p_0(z_0^i) - \sum_{k=1}^K \log \left| \det \frac{\partial f_k}{\partial z_{k-1}} \right|_{z_{k-1}=z_{k-1}^i} \right)$$

$\xrightarrow{\dim z \times \dim z}$

$$\theta = \text{opt\_step}(\nabla_{\theta} \text{Loss})$$

Requirements for  $f_k$ :

- ①  $f_k$ -invertible
- ② fast to compute  
det-jacobian

NICE

Coupling layer  $x = f(z)$ :

$$z = [z_a, z_b]$$

$$z = f^{-1}(x):$$

$$x_a = z_a$$

$$x_b = z_b + NN(z_a, \theta)$$

$$z_a = x_a$$

$$z_b = x_b - NN(x_a, \theta)$$

$$\left| \det \frac{\partial x}{\partial z} \right| = \left| \det \begin{bmatrix} I & 0 \\ \frac{\partial NN}{\partial z_a} & I \end{bmatrix} \right| = 1$$

block:  $\text{input} \rightarrow \text{split} \rightarrow \text{Coupling Layer}(z_a, z_b) \rightarrow$   
 $\rightarrow \text{Coupling Layer}(x_b, x_a) \rightarrow \text{output}$

Flow:  $z \sim \mathcal{N}(z|0, I) \rightarrow \text{block}_1 \rightarrow \text{block}_2 \rightarrow \dots \rightarrow \text{block}_k = x$

Real MVP

$$z = f^{-1}(x):$$

$$z_a = x_a$$

$$z_b = (x_b - t) \odot \exp(-s)$$

$$\det \frac{\partial x}{\partial z} = \det \begin{bmatrix} \mathbb{I} & 0 \\ \frac{\partial t}{\partial z_a} & \text{diag}(\exp(s)) \end{bmatrix} = \prod_{i \in b} \exp(s_i)$$

Affine Coupling Layer

$$z = [z_a, z_b]$$

$$x_a = z_a$$

$$x_b = z_b \odot \exp(s) + t$$

$$s = \text{NN}_s(z_a, \theta)$$

$$t = \text{NN}_t(z_a, \theta)$$

Alow

Batch/Layer/Inst. norm:  $y = s \odot \frac{x - m(x)}{\sigma(x)} + b$

$$(y - b) \odot \sigma^{-1} \odot \sigma(x) + m(x) = x$$

$\frac{\partial y}{\partial x}$  - full matrix

Act Norm:  $y = s \odot x + b$

$$x = (y - b) \odot s^{-1}$$

$$\det \frac{\partial y}{\partial x} = \prod_i s_i$$

Invertible  $1 \times 1$  conv

$$(w, h, c) \rightarrow 1 \times 1 \text{ conv} \rightarrow (\bar{w}, \bar{h}, \bar{c})$$

$$\underset{c \times (hw)}{y} = \underset{c \times c}{W} \underset{c \times (hw)}{x} \Rightarrow x = W^{-1} y$$

$$\det \frac{\partial y}{\partial x} = \det \begin{bmatrix} w & & 0 \\ & w & \\ 0 & \ddots & w \end{bmatrix} =$$
$$= \det W(hw)$$

$$W = P \cdot L \cdot (U + \text{diag}(s))$$

$P$  - permutation matrix  $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ ,  $P^{-1} = P^T, |\det P| = 1$

$L = \begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$ ,  $L_{ii} = 1$

$U = \begin{pmatrix} \circ & & & \\ & \circ & & \\ & & \circ & \\ 0 & & & \circ \end{pmatrix}$

$$\det W = \prod_i s_i$$

block: input  $\rightarrow$  ActNorm  $\rightarrow$  Invertible  $1 \times 1$  conv  $\rightarrow$

$\rightarrow$  Affine Coupling Layer  $\rightarrow$  output

Flow Block: input  $\rightarrow$  block<sub>1</sub>  $\rightarrow$  ...  $\rightarrow$  block<sub>k</sub>  $\rightarrow$  output

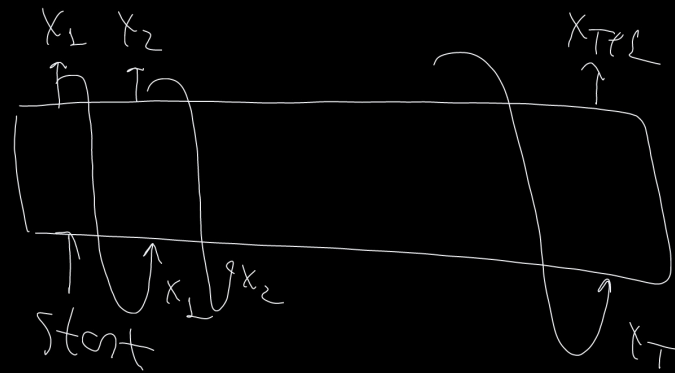
squeeze:  $(w, h, c) \xrightarrow{\text{reshape}} (\frac{w}{2}, \frac{h}{2}, 4c)$

$\overset{\text{image}}{X} \rightarrow \text{squeeze} \rightarrow \text{Flowblock} \rightarrow \overset{z_1}{\text{split}} \left( \begin{array}{l} \text{squeeze} \rightarrow \text{flowblock} \rightarrow \text{split} \rightarrow \dots \\ \rightarrow \text{squeeze} \rightarrow \text{flowblock} \rightarrow z_L \end{array} \right)$

# Inversed Autoregressive Flow (IAF)

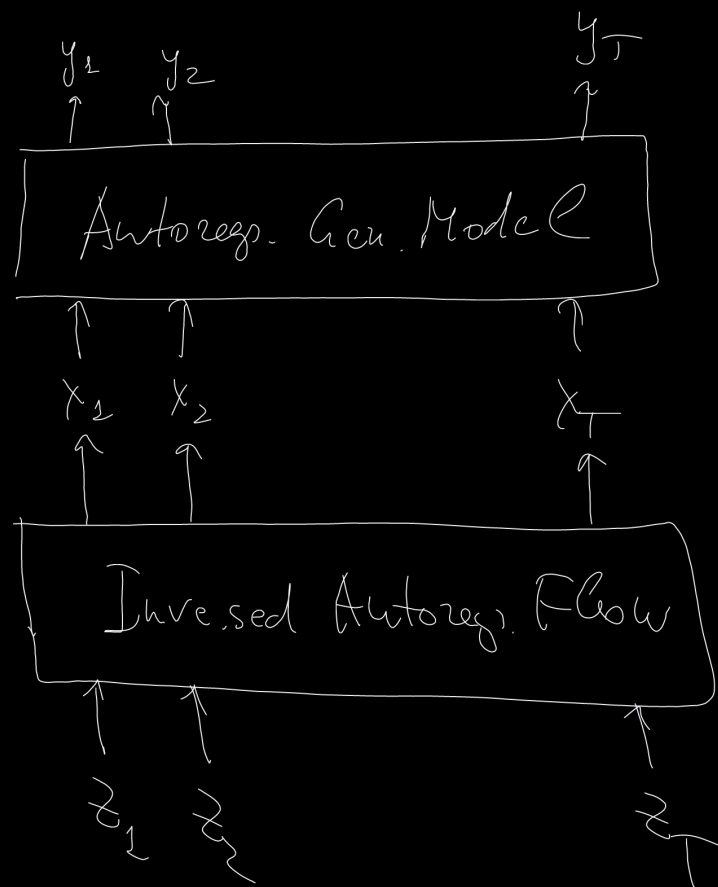
for Distillation of Autoregr. Gen. Models

GPT:



$x \sim p(x)$  — slow

compute  $p(x_{2:T+1} | x_{1:T})$  — fast



$$x \sim p(x) \text{ - slow}$$

$$p(x_{2:T+1} | x_{1:T}) \text{ - fast}$$

$$z_t \sim \mathcal{N}(z(0, I))$$

$$x_t = \sigma(z_{1:t-1}, \theta) z_t + \mu(z_{1:t-1}, \theta)$$

$$x \sim p(x) \text{ fast}$$

$$X_t = \sigma(z_{1:t-1}, \theta) z_t + \mu(z_{1:t-1}, \theta)$$

$$X_1 = \sigma_1 z_1 + \mu_1 \Rightarrow z_1 = \frac{X_1 - \mu_1}{\sigma_1}$$

$$X_2 = \sigma_2 z_2 + \mu_2 \Rightarrow z_2 = \frac{X_2 - \mu_2(z_1)}{\sigma_2(z_1)}$$

⋮

$$\det \frac{\partial X}{\partial z} = \prod_t \frac{\partial X_t}{\partial z_t} = \prod_t \sigma(z_{1:t-1}, \theta)$$

Comp.  $p(x)$   
- slow

$p_T(x)$  - teacher model (GPT, ...)

$p_S(x|\theta)$  - student model (IAF)

$$KL(p_S(x|\theta) \parallel p_T(x)) \rightarrow \min_{\theta}$$

# log-derivativ trick vs. reparameterization trick

$$\mathbb{E}_{p(x|\theta)} f(x) \rightarrow \min_{\theta} \int p(x|\theta) \frac{\nabla_{\theta} p(x|\theta)}{p(x|\theta)} f(x) dx$$

baseline

log-deriv. :  $\nabla_{\theta} \mathbb{E}_{p(x|\theta)} f(x) = \mathbb{E}_{p(x|\theta)} \nabla_{\theta} \log p(x|\theta) (f(x) - b)$

$f$ -arbitrary

$$\text{Loss} = \frac{1}{M} \sum_{j=1}^M \log p(x_j|\theta) (f(x_j) - b),$$

$$\theta = \text{opt\_step}(\nabla_{\theta} \text{Loss}) \quad x_j \sim p(x|\theta)$$

reparam. trick:  $x \sim p(x|\theta) \Leftrightarrow z \sim p(z)$

$$x = g(z, \theta)$$

$g$  - differentiable

Example:  $x \sim \mathcal{N}(x|\mu, \sigma^2)$



$$z \sim \mathcal{N}(z|0, 1), x = \sigma z + \mu$$

$$\begin{aligned} \nabla_{\theta} \mathbb{E}_{p(x|\theta)} f(x) &= \nabla_{\theta} \mathbb{E}_{p(z)} f(g(z, \theta)) = \\ &= \mathbb{E}_{p(z)} \nabla_{\theta} f(g(z, \theta)) \end{aligned}$$

$$\text{Loss} = \frac{1}{M} \sum_{j=1}^M f(g(z_j, \theta)), \quad z_j \sim p(z)$$

$$\theta = \text{opt\_step}(\nabla_{\theta} \text{Loss})$$

requirements: ①  $p(x|\theta)$  is reparameterisable  
 ②  $f, g$  - differentiable

$$KL(p_S(x|\theta) \parallel p_T(x)) =$$

$$= \mathbb{E}_{p_S(x)} \log \frac{p_S(x)}{p_T(x)} = \mathbb{E}_{p_S(x|\theta)} \log p_S(x) - \mathbb{E}_{p_S(x|\theta)} \log p_T(x)$$

$$\mathbb{E}_{p_S(x|\theta)} \log p_S(x|\theta) = \mathbb{E}_{p(z)} \left( \overset{\text{const}}{\log p_0(z)} - \sum_{t=1}^T \log \sigma(z_{1:t-1}, \theta) \right)$$

$$\prod_{t=1}^T \mathcal{N}(z_t | 0, I)$$

$$p_T(x) = p(x_1) p(x_2 | x_1) p(x_3 | x_1, x_2) \dots =$$

$$= \prod_{t=1}^T p(x_t | x_{1:t-1})$$

$$E_{p_S(x)} \log p_T(x) = \sum_{t=1}^T E_{p_S(x)} \log p_T(x_t | x_{1:t-1}) =$$

$$= \{ \text{reparam. trick} \} = \sum_{t=1}^T E_{p(z)} \log p_T \left( \sigma(z_{1:t-1}, \theta) z_t + \mu(z_{1:t-1}, \theta) \right) \Big|_{\{ \sigma \cdot z + \mu \}_{1:t-1}}$$

differentiable!